

Assignment 1

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1.1 Report on the NOAA Arctic Report Card

The following report is based on the National Oceanic and Atmospheric Administration's (NOAA) Arctic Report Card: Update for 2025.

Link: <https://arctic.noaa.gov/report-card/report-card-2025/sea-surface-temperature-2025/>

2 Audience Profile

High school students have diverse and stratified levels of knowledge. In general, one can expect that the average senior-level student is more knowledgeable than the average freshman-level student, but within the levels there is some consistency. Students can be expected to know basic science by the time they begin high school, though they may not yet have a specific background in biology, chemistry, or physics. Therefore, general scientific language and ideas, such as cause and effect, hypothesis, and other similar ideas, should be understandable to the average high school student. However, jargon or theories specific to certain fields would not. At the same time, high school students are on the verge of becoming adults themselves, and should not be treated as if they are unintelligent or ignorant.

It may be effective to invite them into the topic with questions or proposed visualizations for them to picture in their own heads. This could drive engagement and investment with the subject, encouraging them to pay attention and possibly continue their learning on the topic afterwards. Similarly, keeping the language familiar (we, our, etc). They might ask questions such as: why should I care? Or why does that matter? Now what? As such, leading them in another direction where they can learn more and ask more questions, probably in the conclusion, could be valuable.

As the students are young and impressionable, it is especially crucial to not mislead them by over or understating conclusions. Also, the reasons why may be even more important to them than the findings themselves, as these students have their whole lives ahead of them. On the cusp of pursuing long term careers or higher education, audience members are in a position where they could absolutely be the ones addressing technological and societal challenges to humanity in the future.

3 Rewrite

News about climate change can often be extremely inflammatory, using broad statements about the "End of the World!" or the opposite, that there is no problem at all. This dramatized news often

isn't based on outright incorrect data, but rather cherry-picked snippets that, while alone are true, don't tell the whole story. To be an educated consumer of news, a person should try to understand where the data comes from and the stories it tells.

The Arctic Ocean contains a vastly important system for the stability of our earth: sea ice. This system acts like a regulator for the rest of the earth, stabilizing temperatures and ecosystems. The Arctic is very close in proximity to population centers in the northern hemisphere: think North America, Europe, and Asia. Any changes to the Arctic and its sea ice are felt quickly and intensely by huge populations of people, plants, and animals.

Now, imagine two patches of the Arctic sea: one is mostly covered in sea ice, and the other is mostly dark ocean water. The sunlight is shining equally on both patches, but which one will be warmer? Just like how a darker t-shirt gets warmer on a sunny day, the patch of dark ocean water will heat up more than the reflective ice one. Since the patch with mostly dark ocean water heats up more, the bits of ice there also melt faster, revealing more of the dark ocean water, which in turn makes it warm up more. This loop of melting and heating and melting is called ice-albedo feedback, and it happens every summer in what is called the melt season. Eventually, during the autumn, temperatures cool down and there is a refreeze of sea ice. This process itself is natural, but its extent each year is variable.

An important indicator of the ice-albedo feedback process is sea-surface temperature, which, as the name indicates, is how warm the very top surface of the sea is. August is a standard time to look at summer sea-surface temperatures, because the ice should be at a seasonal low. This is because it's generally the warmest time, and any melting feedback loops have had time to run their course before temperatures drop for the autumn and refreezing starts. So, by using the same time in August every year, we can make year-to-year comparisons to see how everything changes over time. We can also make observations of larger trends over decades.

Researchers looked at both the year-to-year and trends over longer time periods. They found that compared to the same time in 2024, certain areas were unusually warm and other areas were unusually cold. What this shows is not necessarily the trend of a warming world, but instead volatility, or the degree to which temperatures can swing back and forth over relatively short times.

Looking at the longer trends paints a more distinct picture. The long-term statistics show consistent warming since 1982 in nearly the entire Arctic Ocean. This ties back to the earlier point about the ice-albedo effect, as the sea-surface temperature has an effect on ice cover and vice versa. Some areas were around 1-4°C warmer than the historical average, while some lesser areas were around 1-2°C colder. Considering both the warmer and cooler areas, the overall temperature shows a trend of increasing compared to historical numbers.

So from the data, there are two things to keep in mind. First, there is volatility in the year-to-year trends. There are areas with warmer temperatures and areas with cooler temperatures. This data may be useful in predicting or explaining yearly events around ice thawing and freezing, but it should not be used or taken as evidence of greater climate trends. The second point, overall temperatures are increasing when compared to previous decades. While that may be concerning, it does not mean the end of the world is imminent and we should give up. Instead, let that evidence be a reason to ask more questions and look for solutions. We should ask if the warming trend is natural or human-influenced, and how it is influenced. From there, we can research solutions based on the data and facts, rather than dramatic headlines.

4 Reflection

The hardest part to simplify for me was in the actual temperatures recorded and how those fit into the overall trends. They use a few different seas within the Arctic Ocean for recording temperatures, and use each throughout the report. Some of the seas have anomalously high temperatures and others low. This information could be conveyed simply to an audience, but so many numbers about unknown places to the audience would likely not be useful. So instead, I had to think about conveying the key findings from that data without over or undersimplifying. Luckily, the figures in the report make the data extremely clear, with a map of areas that were warmer/cooler and for different comparisons, relative to 2024 and relative to 1982-2020.

There are a number of ethical risks when translating technical data, and not all are intentional. Simply misunderstanding the data is a risk, especially when it comes from convoluted processes or is unclear through the tables, figures, etc. While perhaps more of an epistemological value, that understanding of the facts is always a risk when translating information from one place to another. On the other hand, there are more nefarious ethical issues that can arise. Many of them revolve around the motivation to push some kind of narrative that is not necessarily in line with the evidence. That may mean cherrypicking data, over/understating of findings, falsifying data, leaving out context or limitations, and more.

Simplification absolutely runs the risk of changing the meaning, especially when specific numbers are translated into simple language. For instance, consider the statement 1-4 degrees warmer in some places and 1-2 degrees colder in others. That gives the idea that there is some warming and some cooling going on, but if one looks at the map, they'd see the cooling area is localized mostly to one region, while the warming is happening in almost every other region of the Arctic.

One way communicators can avoid misleading is by putting themselves in the position of their audience. From there, they can anticipate how someone might read or hear the information they are presenting and what conclusions could be drawn. If these conclusions are in a reasonable direction, then that's fine, but many unreasonable conclusions are foreseeable. While a communicator cannot, and should not, control the audience, they can anticipate the audience and prepare their delivery in an honest manner.